**Quiz-2; Time: 10 minutes**
**Course Title: Statistics and Probability**
**Course Code: STA102**

**ID:** 2025-1-60

**Q1.** A fair die is rolled. The probability of getting a number less than 4 is:

a) 1/2

b) 2/3

c) 1/3

d) 5/6

**Q2.** If P(A) = 0.6, P(B) = 0.5, and P(A∪B) = 0.9, then P(A∩B) is:

a) 0.1

b) 0.2

c) 0.4

d) 0.5

**Q3.** Which of the following is true for independent events A and B?

a) P(A|B) = P(A)

b) P(A∩B) = 0

c) P(A|B) = P(B)

d) P(A)+P(B) = 1

**Q4.** If a coin is tossed twice, the sample space contains:

a) 2 outcomes

b) 3 outcomes

c) 4 outcomes

d) 6 outcomes

**Q5.** A valid PDF f(x) must satisfy:

a) $f(x) \geq 0$ and $\int_{-\infty}^{\infty} f(x)dx = 1$

b) $f(x) \leq 0$ and sum = 1

c) Mean = 0

d) Variance = 1

**Q6.** Expectation E[X] of a discrete random variable is:

a) $\sum x\, p(x)$

b) $\int_{-\infty}^{\infty} x f(x) dx$

c) $\sum p(x)$

d) $\sum x^2\, p(x)$

**Q7.** If X is a continuous random variable, then $P(X = x)$:

a) Always 1

b) Always 0

c) Between 0 and 1

d) Undefined

**Q8.** If X has PDF $f(x) = 1$ for $0 \leq x \leq 1$, E[X] is:

a) 0.25

b) 0.5

c) 0.75

d) 0.1

**Q9.** What is the expected value E(X) of a random variable X?

a) Mode of X

b) Maximum of X

c) Weighted average of possible values

d) Midpoint of the distribution

**Q10.** Which of the following is always true for a CDF F(x)?

a) F(x) decreases

b) F(x) is constant

c) F(x) is non-decreasing

d) F(x) is always greater than 1